Categroy Theory Reading Group

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There is a vast variety of motivations to study categories which stem from various domains of formal inquiry, so I choose to first give the definition of a category, and then the motivations will come into light once we start examining particular examples of categories.

0.1 Definition of a Category

In what follows we give the definition (i.e. an axiomatization) of what a *category* is. For readability, we phrase it using some natural language, but for those interested, it is easily expressible formally in either two sorted first order logic or in single sorted first order logic with a predicate for each of the two sorts: *objects* and *arrows*. Without further ado:

Definition 1. A *category* consists of the following data:

- **Objects:** $A, B, C, \ldots$
- **Arrows:** $f, g, h, \ldots$
- For each arrow f there are given objects $\mathrm{dom}(f), \mathrm{cod}(f)$, called the *domain* and *codomain* of f. We will write $f : A \to B$ or draw:

  $$A \xrightarrow{f} B$$

  to indicate that $A = \mathrm{dom}(f)$ and $B = \mathrm{cod}(f)$.
- Given arrows $f : A \to B$ and $g : B \to C$, i.e. with $\mathrm{cod}(f) = \mathrm{dom}(g)$, there is given an arrow

  $$g \circ f : A \to C$$

  called the *composite* of f and g, we read $g \circ f$ as "f *after* g". The information given here about the domains and codomains can be summarized by the following diagram:

$$\begin{tikzcd} & B \arrow[dr, "g"] & \\ A \arrow[ur, "f"] \arrow[rr, "g\circ f"'] & & C \end{tikzcd}$$

- For each object A there is given an arrow

  $$1_A : A \to A$$

  called the *identity arrow* of A.

These data are required to satisfy the following laws:

- **Associativity:** $h \circ (g \circ f) = (h \circ g) \circ f$ for all $f : A \to B$, $g : B \to C$, $h : C \to D$.
- **Unit:** $f \circ 1_A = f = 1_B \circ f$ for all $f : A \to B$.

Notice, that the associativity law makes the expression $h \circ g \circ f$ well defined by $h \circ g \circ f := (h \circ g) \circ f = h \circ (g \circ f)$.

Furthermore there is a handy way to express diagrammatically the judgment that some arrows are equal. We say that a diagram *commutes* to express the fact that for any two objets A and B depicted in the diagram, any composition with domain A and codomain B of the arrows depicted is equal to any other such composition. For example to say that the diagram:

$$\begin{tikzcd} & B \arrow[dr, "g"] & \\ A \arrow[ur, "f"] \arrow[rr, "h"'] & & C \end{tikzcd}$$

commutes, is to preciseely to say that $g \circ f = h$

We can then, for example, express the information in the Unit axiom by saying that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "1_A"] \arrow[dr, "f"'] & A \arrow[d, "f"] \arrow[dr, "f"] & \\ & B \arrow[r, "1_B"'] & B \end{tikzcd}$$

0.2 Examples

Besides providing intuition by giving examples, we can also think of this as giving models of category theory. Once again, for those interrested in the formal details, the first example actually gives a second order model since the domain in which objects are interpreted is the *class* of all sets (this is axiomatized by NBG).

0.3 Set

The prototipical but in some sene also least interresting example of a category is the categry of sets and functions **Set**.

- Objects: sets
- Arrows: functions between sets, the domains and codomains are those the domains and codomains in set theoretic sense.
- Composition of Arrows: composition of functions
- Identity Arrows: identity functions

If you are familiar with these notions form set theory, it is easy to see that composition of functions is Associative and the identity functions satisfy the Unit axiom for identity arrows. Thus, **Set** is a category.

Having said so, it is still useful to review the set theoretic definitions of *function* and *composition* to see how awkward they are.

Firstly, a *relation* R between sets A and B is a subset of their cartesian product, i.e. $R \subseteq A \times B = \{(a,b)|\ a \in A \ \ b \in B\}$.

Then a *function* f with domain A and codomain B in set theoretic terms is to be thought of as the set of all input-output pairs, i.e. it is a relation $f \subseteq A \times B$ satisfying the following constraint:

$$\text{For all } a \in A \text{ there exists a unique } b \in B \text{ so that } \ (a,b) \in f$$

This unique b is then denoted as $f(a)$.

The *identity function* of A is defined as:

$$1_A := \{(a,a)|\ a \in A\}$$

Notice however, that the identity relation on A is exactly the same set, i.e. $=_A := \{(a,a)|\ a \in A\}$, thus set theoretically the identity relation and the identity function are the same entity. However conceptually they feel very different. In future session we will see how this is remedied in category theory.

Composition of functions is defined in set theory as follows:

Let A, B, C be sets, and let $f\colon A \to B$ and $g\colon B \to C$ be functions. The *composition* of g with f is the function

$$g \circ f\colon A \to C$$

defined by

$$(g \circ f)(a) := g(f(a)) \quad \text{for all } a \in A.$$

the associativity of composition is then trivial.

However if are to be fully formal, the following akward definition should be stated:

Let $f \subseteq A \times B$ and $g \subseteq B \times C$ be functions. The *composition* $g \circ f$ is the subset of $A \times C$ given by

$$g \circ f = \{(a, c) \in A \times C \mid \exists\, b \in B \text{ such that } (a, b) \in f \text{ and } (b, c) \in g\}.$$

It is not that hard to check then that $g \circ f$ is indeed a function.

0.4 Preorder Categories

We will see how every *preordered set* (*poset*) gives rise to a category.

Definition 2. A *preordered set* is the pair $(P, \leq)$ where P is a set and $\leq \subseteq P \times P$ is a binary relation satisfying for any $a, b, c \in P$:

- (*Reflexivity*) $a \leq a$
- (*Transitivity*) If $a \leq b$ and $b \leq c$, then $a \leq c$

There is then a natural way how $(P, \leq)$ defines a category $\mathbf{P}$ given by:

1. The objects of $\mathbf{P}$ are the elements of P
2. For any $a, b \in P$ there is an arrow $\leq_a^b \colon a \to b$ iff $a \leq b$

Thus between any two objects there are at most one arrow. Transitivity of the partial order makes well defined the compositions $\leq_b^c \circ \leq_a^b = \leq_a^c$, represented by the commutative diagram[1]

$$\begin{tikzcd}
& c \arrow[dl, "\leq_b^c"'] \arrow[dd, "\leq_a^c"] \\
b \arrow[dr, "\leq_a^b"'] & \\
& a
\end{tikzcd}$$

Reflexivity of the partial order gives the identity arrows $\mathrm{id}_a = \leq_a^a$.

Since given any domain and codomain there is at most one arrow between them the associativity and identity axioms are trivially satisfied.

This example shows us that the we are not forced to think of arrows of a category as morphisms or functions in the usual sense, since the arrows in $\mathbf{P}$ do not denote any evident morphisms.

Another big lesson that thee categories give is that instead of starting out with a set theoretic notion of poset and defining a category we might have proceed with categorical notions altogether. In particular:

Definition 3. A category $\mathcal{C}$ is called *posetal* if between any pair of objects there is at most one arrow in any direction.

Thus we might think that it is the posetal categories that give rise to partial orders and not the other way around.

[1]Note that any Hasse diagram is easily transformed into a commutative diagram by assigning the upward direction to each edge

0.4.1 Specifc Preordered Sets

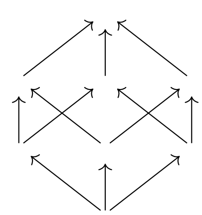
Many preorders of interrest:

- The transitive and reflexive frames in the semantics of modal logic
- Subsets of a given set with the relation $\subseteq$
- Formulas of a logic[2] under (syntactic or semantic) consequence, i.e. $\phi \vdash \psi$, since $\phi \vdash \phi$ and by the cut rule from $\phi \vdash \psi$ and $\phi \vdash \theta$ follows $\phi \vdash \psi$.
- The divisibility relation between integers
- Etc.

Suitably restricted, some of these preorders produce identically-looking categories[3]. For examaple the the positive divisors of 30 of which there are 8 form an identical-looking preordered set as the subsets of a 3 element set $\{\alpha, \beta, \gamma\}$. Indeed if we pair them up like so

$$\begin{array}{rcl}
\varnothing & \longleftrightarrow & 1 \\
\{\alpha\} & \longleftrightarrow & 2 \\
\{\beta\} & \longleftrightarrow & 3 \\
\{\gamma\} & \longleftrightarrow & 5 \\
\{\alpha, \beta\} & \longleftrightarrow & 6 \\
\{\alpha, \gamma\} & \longleftrightarrow & 10 \\
\{\beta, \gamma\} & \longleftrightarrow & 15 \\
\{\alpha, \beta, \gamma\} & \longleftrightarrow & 30
\end{array}$$

Then then one set is a subset of another iff the divisors assigned to them devide one another. Thus their category-theoretic diagrams will both look like



This example shows that when we obtain a category from some mathematical structure, category theoretic means may be insufficient to distinguish from which structure did the category come from. So are category theoretic means enough to distinguish, for example the category **Set**? Surprisingly they are, through the category theoretic notion of universal construction. We begin discussing them in the following sections.

[2]This does not apply to substructural logics, however they were developed by Girard with explicit reference to category theory

[3]In the future this will be defined as an isomorphisms between categories

1 Terminal Objects

In the category **Set** can we tell apart one set form another jut looking at the arrows? It turns out that we can upto the cadinality of sets. For conider finite sets with sizes m and n, then there will be n^m functions from the former to the latter. So in particular if we know that a set T is such a set that each other has exactly one arrow from it to T, then $|T|^m = 1$ and so the size of T can only be 1. In this way we arrive at a categorical charchterisation of *signleton sets*, i.e. sets conatinig one element.

Definition 4 (Terminal Object)**.** An object T is said to be *terminal* in a category $\mathcal{C}$ iff for any object X of $\mathcal{C}$ there exists a unique arrow from X to T.

Terminal objects in **Set** have a special role: they allow us too "look into" what's happening inside of objects. Take any singleton $T = \{*\}$, then there is a natural correspondence between *arrows from* the terminal object and *elements* of a set. Indeed, a function $f : T \to X$ is completely determined by the image of the sole point:

$$f(*) = x \in X.$$

Thus each element $x \in X$ is the same thing as a function $T \to X$.

Therefore, the terminal object allows us to "look inside" a set X by identifying its elements with maps from T.

This is often expressed by saying that:

"global elements of X are morphisms $T \to X$."

So T acts as a *probe* object: to describe an element of X, we describe a map from the terminal object into X.